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SHORT-PAPER

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# Fake News Detection via Multi-scale Semantic Alignment and Cross-modal Attention

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## ABSTRACT

The effective utilization of multimodal information is crucial for the task of fake news detection. The primary challenge lies in reducing the semantic distance between the news evidences of different modalities and accurately aligning them in the latent space. It has been observed that the relevance of news image content varies across coarse-to-fine scales, depending on the object and its class label. For instance, the visual evidence to justify a news item sometimes may be a very local detail such as the tiny changes of facial expression and limb position while sometimes may be the global composition. Consequently, how to align news text with image at the most discriminative scale significantly impacts detection performance. However, very few studies have addressed this issue in fake news detection. In this paper, we delve deeper into this issue and propose a simple yet effective Multi-scale Semantic Alignment and Cross-modal Attention (MSACA) network. Specifically, we construct hierarchical multi-scale images for each news data, enhance the semantic consistency between text and images in the latent space, and employ an attention module to select the deterministic embeddings in an end-to-end manner. Extensive experiments on two real-world benchmarks demonstrate the superior performance of our proposed MSACA network.

## CCS CONCEPTS

• **Applied computing** → *Sociology*.

## KEYWORDS

Fake news detection, Multi-scale representation, Multimodal learning

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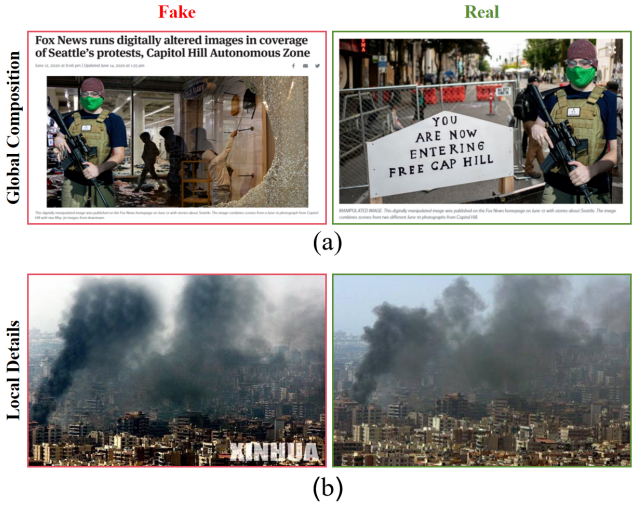


**Figure 1: Two typical cases of fake news where the visual evidence of (a) is about the global composition while justifying (b) needs to focus on local details.**

## 1 INTRODUCTION

With the widespread adoption of smart devices and the Internet connectivity, people are increasingly turning to social platforms as their primary source of information. However, this trend also amplifies the proliferation and dissemination of fake news [19], posing a significant threat to social harmony and stability.

To enable efficient early detection of fake news, various approaches have been proposed for multimodal fake news detection. The transformer-based models, *i.e.*, Vilt [7] and MBT [9], fuse the multimodal features and feed them into downstream model for detection. However, these models often come with high computational costs. A more prevalent approach is to utilize multimodal data as input for pre-trained models, extracting high-level features for inference [1, 4, 11, 17, 18, 21, 23, 25].

It has been observed that semantic inconsistencies between different modalities are one of the most significant problems. Existing

work [1] reduces the semantic gap between modalities via VAE-based semantic alignment. CSFND [10] fill the semantic gap by introducing the context information into the representation learning process. However, as images at different scales focus on different aspects of the visual concept [26], the news text data may be not always perfectly aligned with the image at a fixed resolution. To illustrate, as shown in Figure 1, the visual evidence to justify a news sometimes may be a very local details such as the tiny changes of facial expression or limb position while sometimes can be the global composition. Therefore, fully leveraging the spatial scale information of image contents is crucial for overall detecting performance.

To address this problem, we propose the Multi-scale Semantic Alignment and Cross-modal Attention (MSACA) network, as shown in Figure 2. MSACA consists of two components: i) the Multi-scale Semantic Alignment (MSA) module, which aligns the distributions of each modality from coarse to fine, and ii) the Scale-wise Cross-modal Attention (SCA) module, which selects the most discriminative scale-wise uni-modal and cross-modal embeddings in the latent space. By combining MSA and SCA, our pipeline is capable of matching news text with the corresponding image at the most discriminative scale to make accurate predictions. Extensive experiments on two real-world benchmarks demonstrate the superior performance of our MSACA.

The main contributions of this paper can be summarized as follows:

- We are the first to introduce a multi-scale pipeline to reduce the semantic ambiguities between different modalities and improve the quality of cross-modal representations.
- We further propose the scale-wise cross-modal attention to generate weights for available uni-modal and multimodal news embeddings, thereby selecting the most discriminative representations to make the final prediction.
- Extensive experiments on two real-world benchmarks demonstrate the effectiveness of MSACA in distinguishing fake news compared with other methods.

## 2 RELATED WORKS

In the early stage, researchers primarily experimented with new pre-trained models to replace the traditional uni-modal feature extraction models. Jin et al. [4] extracted image and text features via VGG-19 and LSTM models. Sinhal et al. [17] replaced the traditional recurrent neural network with the BERT model. For the complementarity of multimodal information, EM-FEND [12] used an optical character recognition model to extract important information from news images. For the consistency of multimodal information. Song et al. [18] selectively learned information from one modal to another based on a bi-directional cross-attention mechanism and combined it with a residual network to preserve the original features. MVAE [6] proposed learning the latent distribution of fake news features using a variational autoencoder. Diachanou et al. [2] computed the extracted uni-modal features by cosine similarity to obtain modal consistency features, and finally concatenated the three parts of the features for the final prediction. CAFE [1] combined the VAE and the KL divergence to learn semantic ambiguity across different modalities. Inspired by CLIP [14], Zhou et al. [27] obtained the cross-modal fusion vectors via cross-modal similarity learning. CMC [22] and COOLANT [20] learned the consistency of cross-modal features using contrastive learning methods and fine-tuned the classification model with cross-modal features.

To sum up the above work, although it has yielded certain results, it has largely overlooked the significance of multi-scale visual representation for detection accuracy. In this paper, we aim to address this gap by effectively leveraging MSA and SCA modules to enhance detection performance.

## 3 APPROACH

The task of fake news detection is a type of anomaly detection. Given a news data $x = \{t, v, y\}$ where $t$ represents text and $v$ refers to visual content *i.e.,* image, the objective of this task is to find a mapping to determine if the news $N$ is true ($y = 0$) or false ($y = 1$).

In this paper, we propose the so-called Multi-scale Semantic Alignment and Cross-modal Attention (MSACA) network, as shown in Figure 2, to improve detection performance. We demonstrate the details using a 3-scale MSACA pipeline.

For fair comparison, we choose the same pre-trained model as CAFE [1]. Specifically, for a given news data tuple $x_i = \{t_i, v_i, y_i\}$ where $v_i \in \mathbb{R}^{224\times224\times3}$, we first downsample $v_i$ to get the multi-scale visual inputs $\{v_i^s\}_{s=1}^3$, then extract the uni-modal features for $t_i$ and $\{v_i^s\}_{s=1}^3$ via pre-trained BERT [5] and ResNet-34 [3] respectively.

$$\boldsymbol{\phi}_i^t = \text{BERT}(t_i), \quad \boldsymbol{\phi}_i^{(v,s)} = \text{ResNet-34}(v_i^s), \tag{1}$$

Given the inherent semantic ambiguity present across two modalities, $\boldsymbol{\phi}_i^t \in \mathbb{R}^{N\times768}$ and $\boldsymbol{\phi}_i^{(v,1)}, \boldsymbol{\phi}_i^{(v,2)}, \boldsymbol{\phi}_i^{(v,3)} \in \mathbb{R}^{N\times512}$ are fed into two VAE modules to perform semantic alignment, where N represents the number of news items. We denote the textual and visual VAE encoders and decoders as $f_e^t$, $f_d^t$, $f_e^v$ and $f_d^v$ respectively.

$$\begin{aligned} &z_i^t \sim N(\mu_i^t, \sigma_i^t),\ z_i^{(v,s)} \sim N(\mu_i^{(v,s)}, \sigma_i^{(v,s)}), \\ &\text{s. t.} \quad \{\mu_i^t, \sigma_i^t\} = f_e^t(\boldsymbol{\phi}_i^t),\ \{\mu_i^{(v,s)}, \sigma_i^{(v,s)}\} = f_e^v(\boldsymbol{\phi}_i^{(v,s)}) \end{aligned} \tag{2}$$

Following this, $z_i^t$ and $z_i^{(v,s)}$ are fed into modal fusion stage to obtain the inter-modal features $z_i^{t\leftarrow(v,s)}$ and $z_i^{(v,s)\leftarrow t}$:

$$\begin{aligned} z_i^{t\leftarrow(v,s)} &= \text{softmax}\left(z_i^t \cdot \left[z_i^{(v,s)}\right]^T / \sqrt{dim}\right), \\ z_i^{(v,s)\leftarrow t} &= \text{softmax}\left(z_i^{(v,s)} \cdot \left[z_i^t\right]^T / \sqrt{dim}\right), \end{aligned} \tag{3}$$

As $z_i^{t\leftarrow(v,s)}$ and $z_i^{(v,s)\leftarrow t}$ are already aligned, we directly fuse them through the element-wise product to obtain the final cross-modal features $z_i^{(c,s)}$ at the $s$-th scale:

$$z_i^{(c,s)} = z_i^{t\leftarrow(v,s)} \cdot z_i^{(v,s)\leftarrow t}, \tag{4}$$

**Scale-wise Cross-modal Attention.** In order to select most discriminative uni-modal and cross-modal embeddings for fake news detection, we further propose the SCA module to assign attentions for modal-wise embeddings at each scale, respectively.

Specifically, the available features aligned in the high-dimensional space are firstly mixed with the unaligned features via a residual

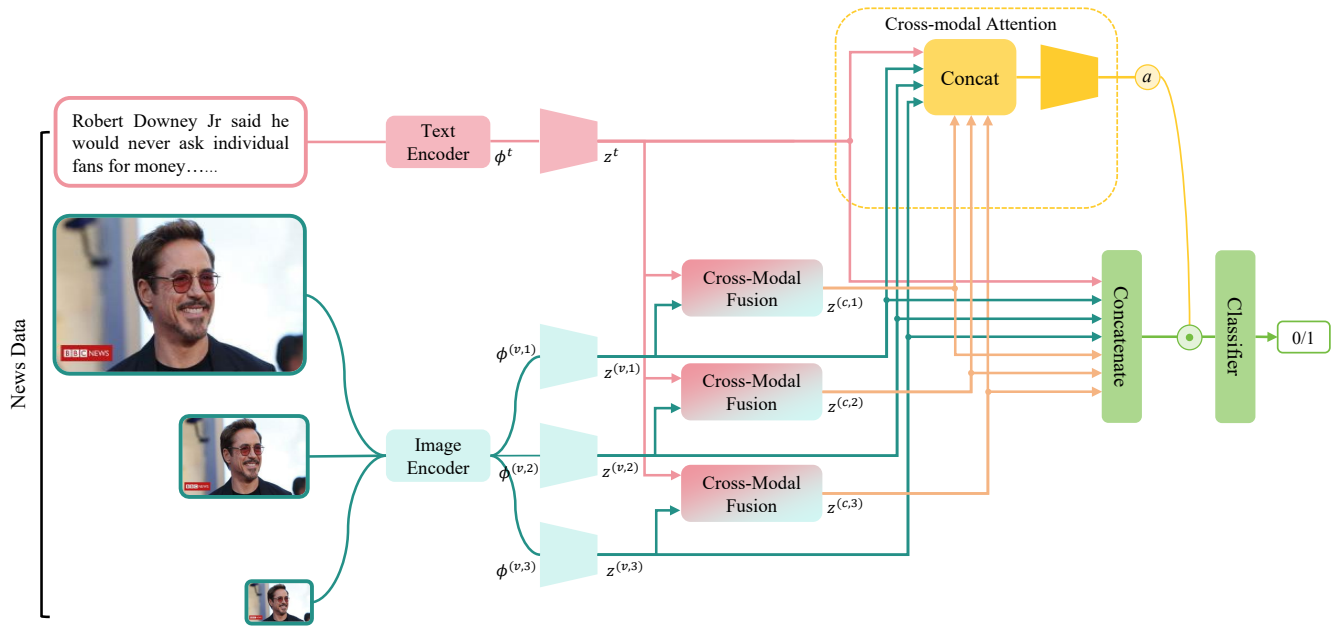


**Figure 2: The overall pipeline of our proposed MSACA network.**

connection as follows:

$$\boldsymbol{\psi}_i^t = z_i^t W_i^t + \boldsymbol{\phi}_i^t, \tag{5}$$

$$\boldsymbol{\psi}_i^{(v,s)} = z_i^{(v,s)} W_i^{(v,s)} + \boldsymbol{\phi}_i^{(v,s)}, \tag{6}$$

$$\boldsymbol{\psi}_i^{(c,s)} = z_i^{(c,s)} W_i^{(c,s)} + \boldsymbol{\phi}_i^{(c,s)}, \tag{7}$$

where $W$ refers to a single-layer mapping to align $\boldsymbol{\phi}$ and $z$.

Then we employ a MLP-based module $h(x;\boldsymbol{\theta}_h)$ to generate the scale-wise attentions:

$$a_i = h(\mathrm{cat}(\boldsymbol{\psi}_i^t, \{\boldsymbol{\psi}_i^{(v,s)}\}^3, \{\boldsymbol{\psi}_i^{(c,s)}\}^3)), \tag{8}$$

By assigning the scale-wise attention to different uni- and cross-modal features, the network is capable of using the most discriminative feature combination to make the final prediction

The prediction label is obtained by multiplying the attention weights through the element product with the uni-modal and cross-modal embeddings before entering the classifier, The final prediction label can be expressed as:

$$\hat{y}_i = g(\mathrm{cat}(\boldsymbol{\psi}_i^t, \{\boldsymbol{\psi}_i^{(v,s)}\}^3, \{\boldsymbol{\psi}_i^{(c,s)}\}^3) \cdot a_i), \tag{9}$$

Lastly we use cross-entropy loss $\mathcal{L}_{cls}$ to train the model parameters:

$$\mathcal{L}_{cls} = \frac{1}{N}\sum_i y_i log(\hat{y}_i) + (1-\hat{y}_i) log(1-y_i). \tag{10}$$

**Multi-scale Semantic Alignment.** This module aims at improving the semantic consistency between textual and visual data of each news, which helps to boost the overall detect accuracy.

To demonstrate, we first select the pairs of real news $\{x_j, y_j | y_j = 0\}_j^{N_R}$ where $N_R$ is the total number of real news data in training set, and generate their specific posterior distributions $z_j^t$ and $z_j^{(v,s)}$ as Eq. 2. Then we measure the KL divergence between the original and VAE embedding distributions, and minimize it to improve the cross-modal semantic consistency.

$$\mathcal{L}_{MSA} = \frac{1}{N_R}\sum_j (\mathrm{KL}(N(\mu_j^t, \sigma_j^t) || N(0,1)) + \sum_s \mathrm{KL}(N(\mu_j^{(v,s)}, \sigma_j^{(v,s)}) || N(0,1)) ), \tag{11}$$

The overall loss for MSACA is defined as following:

$$\mathcal{L} = \mathcal{L}_{CLS} + \alpha \mathcal{L}_{MSA}, \tag{12}$$

where $\alpha$ is a hyper-parameter to control the impact of MSA during training.

## 4 EXPERIMENTS AND RESULTS

### 4.1 Datasets

We conduct experiments on two widely used multimodal fake news detection datasets.

**Weibo** dataset [4] was collected from the Chinese social media platform, Weibo. The fake news in the dataset was verified by the debunking system from May 2012 to January 2016. Following [13, 21], we removed the news items with videos and those without texts or images, and divided the remaining data into training and testing sets with a ratio of 8:2.

**GossipCop** dataset, released by Shu et al.[15], was collected from the GossipCop website. Following [16, 24], we manually removed logos from the articles and dropped samples that either lacked images or contained GIFs. The rest of the data was divided into the training set and the test set at a ratio of 8:2.

**Table 1: Performance comparison among different models on Weibo and GossipCop Datasets.**

| Method | | Weibo | | | | | | | GossipCop | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Acc. | Fake News | | | Real News | | | Acc. | Fake News | | | Real News | | |
| | | | Pre. | Rec. | F1 | Pre. | Rec. | F1 | | Pre. | Rec. | F1 | Pre. | Rec. | F1 |
| BERT | [5] | 0.844 | 0.883 | 0.806 | 0.843 | 0.808 | 0.885 | 0.845 | 0.850 | 0.689 | 0.259 | 0.546 | 0.891 | 0.947 | 0.905 |
| ResNet-34 | [3] | 0.664 | 0.630 | 0.856 | 0.726 | 0.746 | 0.456 | 0.566 | 0.804 | 0.487 | 0.206 | 0.289 | 0.833 | 0.948 | 0.887 |
| SAFE | [4] | 0.762 | 0.831 | 0.724 | 0.774 | 0.695 | 0.811 | 0.748 | 0.838 | 0.758 | **0.558** | 0.643 | 0.857 | 0.937 | 0.895 |
| att-RNN | [4] | 0.772 | 0.854 | 0.656 | 0.742 | 0.720 | 0.889 | 0.795 | 0.838 | 0.657 | 0.423 | 0.509 | 0.817 | 0.859 | 0.884 |
| MVAE | [6] | 0.824 | 0.854 | 0.769 | 0.809 | 0.802 | 0.875 | 0.837 | 0.861 | 0.708 | 0.491 | 0.580 | 0.882 | 0.961 | 0.916 |
| EANN | [21] | 0.827 | 0.847 | 0.812 | 0.829 | 0.807 | 0.843 | 0.825 | 0.864 | 0.702 | 0.518 | 0.594 | 0.887 | 0.956 | 0.920 |
| CAFE | [1] | 0.840 | 0.855 | 0.830 | 0.842 | 0.825 | 0.851 | 0.837 | 0.867 | 0.732 | 0.490 | 0.587 | 0.887 | 0.957 | 0.921 |
| SpotFake | [17] | 0.869 | 0.877 | 0.859 | 0.868 | 0.861 | 0.879 | 0.870 | 0.856 | 0.715 | 0.386 | 0.487 | 0.863 | 0.935 | 0.902 |
| SpotFake+ | [16] | 0.870 | 0.887 | 0.849 | 0.868 | 0.855 | 0.892 | 0.873 | 0.858 | 0.732 | 0.372 | 0.494 | 0.866 | 0.962 | 0.914 |
| HMCAN | [13] | 0.885 | 0.920 | 0.845 | 0.881 | 0.856 | 0.926 | 0.890 | 0.872 | 0.759 | 0.395 | 0.603 | 0.886 | 0.958 | 0.919 |
| MSACA *w\o MSA.* | | 0.876 | 0.912 | 0.841 | 0.875 | 0.842 | 0.913 | 0.876 | 0.861 | 0.801 | 0.497 | 0.603 | 0.871 | 0.943 | 0.905 |
| MSACA *w\o SCA.* | | 0.881 | 0.907 | 0.859 | 0.882 | 0.856 | 0.905 | 0.880 | 0.864 | 0.806 | 0.501 | 0.615 | 0.874 | 0.946 | 0.910 |
| MSACA | | **0.903** | **0.935** | **0.873** | **0.903** | **0.872** | **0.935** | **0.902** | **0.887** | **0.816** | 0.538 | **0.648** | **0.897** | **0.971** | **0.933** |

**Table 2: Performance comparison of different scales in MSA module on Weibo Dataset.**

| | 1-scale | 2-scale | 3-scale | 4-scale | 5-scale |
|---|---|---|---|---|---|
| Acc. | 0.892 | 0.898 | 0.903 | 0.903 | 0.904 |
| F1 | 0.891 | 0.899 | 0.902 | 0.903 | 0.904 |

## 4.2 Implementation Details

For news text, we set the length limitation of input text to 200 words for Weibo and 512 words for GossipCop. For news images, we resized images to 3-scale (*i.e.* 224x224, 112x112 and 56x56) resolutions and fed them to a pre-trained ResNet-34. For the VAE module, we implemented the modal-wise MLP heads with three fully-connected layers with 64 hidden units in each layer. The hyper-parameter $\alpha$ was set to 0.1. The Adam [8] solver was used to train the model for 100 epochs with a batch size of 64 and an initial learning rate of 5e-4. All experimental results reported are the averaged scores of five runs with different random seeds. The model was trained on multiple NVIDIA TESLA V100 GPUs.

## 4.3 Results and Analysis

Table 1 shows the experimental results of our proposed MSACA. On Weibo, our proposed method MSACA outperforms all the other baselines on all the metrics. From the results, we can observe the importance of reducing the semantic ambiguities between different modalities and the effectiveness of scale-wise cross-modal attention for modal embeddings. The multimodal method is generally superior to the uni-modal method, which means extracting multimodal features from both modalities is essential in detecting fake news. On GossipCop, it is seen that the performance of MSACA outperforms all the other methods on all the metrics. The multimodal methods receive higher accuracy scores than the uni-modal methods, which means learning multimodal features can benefit fake news detection.

## 4.4 Ablation Study

To justify the effectiveness of the different components in MSACA, we conducted comprehensive ablation experiments. The results are reported in the bottom part of Table 1. The compared variants of MSACA are implemented as follows: MSACA *w\o MSA.* is the variant that excludes the Multi-scale Semantic Alignment module. MSACA *w\o SCA.* is the variant that excludes the Scale-wise Cross-modal Attention module. From the results, it can be observed that the detection efficacy of MSACA is superior to that of MSACA *w\o MSA.*, which strongly proves the importance of semantic alignment between modalities of multimodal news data prior to formal detection. The results of the MSACA *w\o SCA.* variant highlight the necessity of applying different attention weights to multi-scale modalities, and prove the importance of adaptively selecting the most discriminative modalities and scales.

To delve deeper into the impact of different scales on detection results, we conducted relevant ablation experiments. As depicted in Fig 2, a significant improvement in model effectiveness is observed when the number of scales is small ( *e.g.*, 1 or 2). However, as the number of scales increases further, the improvement becomes less pronounced. Nevertheless, it's crucial to note that computational complexity and model training time are positively correlated with the number of scales. Therefore, considering these factors, we opted for three scales of visual embedding as the final input for our model.

## 5 CONCLUSIONS

In this paper, we introduce a novel multimodal fake news detection model called Multi-scale Semantic Alignment and Cross-modal Attention (MSACA). Our model addresses the challenge of aligning distributions across different modalities from coarse to fine levels and effectively selecting the most discriminative scale-wise embeddings. Experimental results on two benchmarks demonstrate the role of MSACA in the task of fake news detection.

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